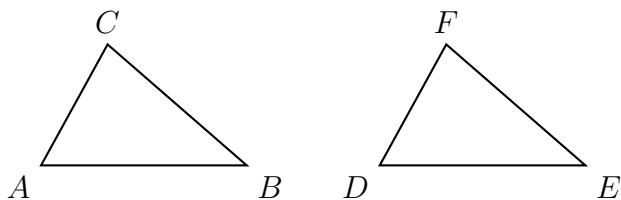


Corresponding Parts of Congruent Triangles

Directions

- A congruence statement lists vertices in matching order. In $\triangle ABC \cong \triangle DEF$, vertex A matches D , B matches E , and C matches F ; side $\overline{AB}$ matches $\overline{DE}$.
- Corresponding parts of congruent triangles are congruent, so matching sides have equal lengths and matching angles have equal measures.
- Set the two matching measures equal, solve, then answer the question that was asked. Show your work in the space provided and include the unit whenever the problem asks for a measurement.



How every congruence statement on this sheet is labelled: matching vertices sit in matching positions. No values are shown — use the numbers given in each problem.

1. A machine shop cuts two congruent steel gusset plates, so that $\triangle ABC \cong \triangle DEF$. The shop drawing lists $AB = (2x + 3)$ cm and $DE = 15$ cm. Find x .

Answer: _____

2. The two spars of a kite frame form congruent triangles with $\triangle PQR \cong \triangle STU$. The pattern lists $m\angle P = (5x - 10)^\circ$ and $m\angle S = 65^\circ$. Find x .

Answer: _____

3. A footbridge is stiffened by two congruent gusset triangles with $\triangle JKL \cong \triangle MNP$. The fabrication drawing gives $JK = (3y + 4)$ ft and $MN = (5y - 12)$ ft.

(a) Find y .

Answer: _____

(b) Find the length of $\overline{MN}$, in feet.

Answer: _____ ft

4. A roof frame is built from two congruent trusses with $\triangle ABC \cong \triangle XYZ$. The plan gives $m\angle A = 48^\circ$, $m\angle B = (4x + 6)^\circ$, and $m\angle Y = 74^\circ$.

(a) Find x .

Answer: _____

(b) Find $m\angle Z$, in degrees.

Answer: _____

5. A sailmaker cuts two congruent triangular panels with $\triangle RST \cong \triangle UVW$. The cutting list gives $RS = 12$ m, $ST = (2n + 5)$ m, $RT = 20$ m, and $VW = 17$ m.

(a) Find n .

Answer: _____

(b) Find the perimeter of $\triangle UVW$, in metres.

Answer: _____ m

6. A bicycle frame joint is reinforced with two congruent plates, $\triangle GHJ \cong \triangle KLM$, with $m\angle G = (3x + 12)^\circ$ and $m\angle K = (5x - 18)^\circ$.

(a) Find x .

Answer: _____

(b) Find $m\angle G$, in degrees.

Answer: _____

(c) Explain how you know that $m\angle J = m\angle M$ without measuring either angle.

7. Two congruent triangular tent panels satisfy $\triangle ABC \cong \triangle DEF$. The pattern lists $AB = (4x - 7)$ cm, $DE = (2x + 9)$ cm, $BC = (3y + 1)$ cm, and $EF = (y + 15)$ cm.

(a) Find x .

Answer: _____

(b) Find y .

Answer: _____

(c) Find the length of $\overline{DE}$, in centimetres.

Answer: _____ cm

8. A crane rests on two congruent support triangles with $\triangle PQR \cong \triangle XYZ$. The specification gives $PQ = (2w + 5)$ m, $XY = 19$ m, $m\angle Q = (7k - 4)^\circ$, and $m\angle Y = 66^\circ$.

(a) Find w .

Answer: _____

(b) Find k .

Answer: _____

(c) The engineer claims that knowing only $PQ = XY$ and $m\angle Q = m\angle Y$ would already guarantee the two triangles are congruent. Explain why that is not enough, and name one additional measurement that would settle it.